

Cost Analysis & Cloud Platform Comparison

One-Year Operating Budget — 14 Hostels, 28 IoT Nodes

WasteWise

IoT-Based Real-Time Food Waste Monitoring System

An initiative supported by

IIT Guwahati — Technology Innovation Hub (TIH)

Prepared for	Detail
Deployment scope	14 hostels · 2 weighing nodes per hostel · ~6,300 students
Period costed	12 months (Year 1 CAPEX + OPEX, and Year 2+ recurring)
Platforms compared	ThingSpeak (paid) vs Amazon Web Services vs self-hosted VPS
Currency basis	Indian Rupees (₹), converted at ₹87 = US\$1

1. Scope and Assumptions

This document budgets one full year of operating WasteWise across all fourteen IIT Guwahati hostels, and compares the two cloud platforms under consideration. Every figure below follows from the assumptions in this section; changing an assumption changes the totals, so they are stated explicitly rather than buried.

Parameter	Value	Note
Hostels	14	Full campus rollout
Nodes (smart scales)	28	Minimum 2 per hostel
Students covered	~6,300	~450 residents per hostel
Message model	1 message per disposal event	Weight, hostel code and Wi-Fi signal sent together
Operating days	365	On a ~280-day academic calendar, volumes fall ~23%
Connectivity	Campus Wi-Fi	Assumed free; 4G SIMs would add ~₹200/node/month
Exchange rate	₹87 = US\$1	Adjust if the rate moves materially

1.1 Data volume — the number that drives every cloud quote

Cloud platforms bill primarily on message count, so the disposal-event rate determines the entire cloud line. Three scenarios are modelled; the Base case is used for all budgets that follow.

Scenario	Events / node / day	Messages / year
Low	200	2.04 million
Base (used throughout)	400	4.09 million
High	800	8.18 million

Even the High scenario stays well inside one ThingSpeak unit (33 million) and is trivial for AWS, so the platform choice is not constrained by capacity — it is decided almost entirely by price and by which features are actually used.

2. Key Finding

HEADLINE ThingSpeak's free tier cannot carry this deployment, and its paid tier costs roughly fifty times more than AWS for the same volume. At Base load, AWS data charges are about US\$12 per year against US\$650 per year for ThingSpeak Standard.

Two independent factors rule out the ThingSpeak free tier:

- Capacity — the free tier allows 3 million messages per year. The Base case consumes 4.09 million.
- Licensing — the free tier is restricted to non-commercial use. The moment sponsor advertisements are displayed, the deployment becomes commercial, which also places the Home and (arguably) Academic licences out of scope.

Why the gap is so large. A ThingSpeak licence bundles data storage together with hosted dashboards, charts and MATLAB analytics. WasteWise does not use any of them — the project already has its own Node.js backend, its own analytics and its own bento dashboard, and reaches ThingSpeak only as a time-series table through a server-side proxy. Paying commercial rates for a feature set that is bypassed is the single largest avoidable cost in this budget.

3. Cloud Platform Comparison

Criterion	ThingSpeak (Standard)	AWS (IoT Core + DynamoDB)	Self-hosted VPS
Data cost / year	US\$650 (₹56,550)	US\$12 (₹1,050)	included in VPS
App hosting / year	Vercel free tier	Lightsail US\$60 (₹5,220)	US\$70 (₹6,000)
Cloud subtotal / year	₹56,550 and up	₹5,200 – 6,500	₹6,000
Year-1 free allowance	None — paid from day one	500k messages/month for 12 months (covers Base entirely)	Not applicable
Message cap	33 million/yr per unit	None — pay per use	None
Update rate limit	1/sec paid; 15 sec free	None	None
Channels	250 per unit	Not applicable	Not applicable
Built-in dashboards	Yes — but unused	No — already built	No — already built
4-channel workaround	Needed on free tier only	Eliminated	Eliminated
Integration effort	Lowest	Highest	Medium
Operational burden	None	Low	Patching and backups are yours
Scales to other campuses	Limited by units purchased	Effortless	Requires a larger instance
Vendor lock-in	High (MathWorks)	Medium	None

Architectural bonus. The current design splits the 14 hostels across four ThingSpeak channels purely to work around the free tier's 15-second write limit. On AWS, on a self-hosted database, or even on paid ThingSpeak, that workaround disappears entirely — removing the channel-routing logic from both the firmware and the server. The migration therefore simplifies the system as well as reducing its cost.

4. Hardware Capital Expenditure

Identical across all three platform options. Prices are Indian-market estimates for the components already proven in the prototype.

Component	Unit price	Qty	Total
ESP32 development board	₹450	28	₹12,600
HX711 24-bit amplifier	₹120	28	₹3,360
Load cell + mounting hardware	₹1,100	28	₹30,800
5 V 2 A power adapter	₹200	28	₹5,600
IP54 enclosure + glands	₹350	28	₹9,800
Wiring, connectors, fasteners	₹180	28	₹5,040
Spares and attrition (~15%)	—	~4 nodes	₹10,080
HARDWARE CAPEX TOTAL			≈ ₹77,000

Per-node cost works out to roughly ₹2,400. A realistic range is ₹60,000 to ₹1,00,000 for the full set, driven almost entirely by load-cell quality.

5. One-Year Budget — Three Options

Excludes television displays, which are costed separately in section 6 because they are optional and revenue-generating rather than operational.

Line item	ThingSpeak	AWS	Self-hosted
Hardware (one-time)	₹77,000	₹77,000	₹77,000
Cloud data / messages	₹56,550	₹1,050 (₹0 in yr 1)	—
Application hosting	₹0 (Vercel)	₹5,220	₹6,000
Domain (.in / .com)	₹1,200	₹1,200	₹1,200
Ad media (Cloudinary free tier)	₹0	₹0	₹0
Maintenance and recalibration	₹10,000	₹10,000	₹10,000
Contingency (10%)	₹14,475	₹9,447	₹9,420
YEAR 1 TOTAL	≈ ₹1,59,225	≈ ₹1,03,917	≈ ₹1,03,620
Year 2+ recurring	≈ ₹74,600	≈ ₹19,000	≈ ₹19,000

SAVING Choosing AWS over ThingSpeak saves approximately ₹55,000 every year in recurring cost — enough to fund roughly twenty additional sensor nodes annually, or to absorb the entire maintenance budget several times over.

6. Advertising Infrastructure and Revenue

The advertisement module allows the deployment to fund itself. It requires a display in each mess, which is the largest single capital line in the project.

Item	Unit price	Qty	Total
32-inch smart television	₹14,000	14	₹1,96,000
Media player (Raspberry Pi / streaming stick)	₹3,500	14	₹49,000
Wall mounts and cabling	₹1,500	14	₹21,000
DISPLAY CAPEX TOTAL			≈ ₹2,66,000

A dedicated media player is recommended over the television's built-in browser, which is typically slow and unreliable for a page that must run unattended for months.

6.1 Revenue potential (illustrative)

Fourteen screens reach approximately 6,300 students three times a day in a captive, high-dwell-time setting — an attractive proposition for local cafes, stationery shops, coaching centres and food-delivery brands.

Scenario	Sponsors × monthly rate	Annual revenue
Conservative	3 × ₹3,000	₹1,08,000
Moderate	5 × ₹4,000	₹2,40,000

Even the conservative case covers the ₹19,000 annual recurring cost on AWS roughly five times over, and repays the display investment in about two years (approximately eleven months in the moderate case).

RECOMMENDED APPROACH Phase the displays. Install televisions in three or four hostels first, sign one paying sponsor, and use the realised rate to validate the revenue model — then expand using advertising income rather than grant funding.

7. Recommendation

Migrate to AWS (or a self-hosted VPS) and retire the ThingSpeak dependency.

The project has already built the two capabilities a ThingSpeak licence provides: a data-ingestion API and an analytics dashboard. Paying ₹56,550 a year for a time-series table that is queried exclusively through the project's own proxy cannot be justified when the equivalent AWS service costs about ₹1,050. The migration also removes the four-channel workaround and the fifteen-second write limit, so it is a genuine architectural simplification and not merely a cost reduction.

7.1 The case for staying on ThingSpeak

One circumstance would justify remaining: if the IIT Guwahati campus-wide MathWorks licence already covers ThingSpeak Academic, the year-one cloud cost falls to zero and it becomes the pragmatic choice for the pilot. This should be confirmed with the institute software cell before any migration work begins.

IMPORTANT CAVEAT Displaying paid sponsor advertisements is commercial use, which the Free, Home and (arguably) Academic licences do not permit. Should the advertising module go live, a Standard licence would be required — so an Academic licence is a short-term answer for the pilot, not a long-term platform decision.

8. Budget Risks

8.1 Advertisement bandwidth

Fourteen screens replaying video every few minutes can generate several terabytes of outbound traffic per year. Left uncached this could exceed US\$450 annually in content-delivery charges — quietly becoming the largest cloud line in the budget. Advertisement media must therefore be cached on the media player or pre-downloaded locally, so that network traffic occurs only when a campaign actually changes.

8.2 Load-cell quality and drift

The ₹1,100 per-cell estimate assumes a reasonable 50–200 kg unit. Inexpensive cells drift and require frequent recalibration, which is why a ₹10,000 annual maintenance line is included. This is a genuine operating cost, not padding: an uncalibrated cell produces implausible readings that undermine confidence in the entire dataset.

8.3 Connectivity

The budget assumes campus Wi-Fi reaches every mess at no cost. Where coverage is absent, a 4G module and data plan would add roughly ₹200 per node per month — approximately ₹67,000 a year across 28 nodes, which would materially change these totals.

9. Basis of Estimates

In the interest of accuracy, the following distinguishes verified figures from estimates.

Figure	Status
ThingSpeak: 1 unit = 33M messages/yr, 250 channels, 1-second updates	Verified on the MathWorks pricing pages
ThingSpeak free tier: 3M messages/yr, non-commercial only	Verified
ThingSpeak Standard: US\$650 per unit per year	Secondary sources — the official page currently shows "purchase options temporarily unavailable". Obtain a written quote.
ThingSpeak Academic pricing	Quote-only; not published
AWS free tier: 500k messages + 2.25M connection-minutes/month for 12 months	Verified

Figure	Status
AWS messaging US\$1.00/million; connectivity US\$0.08/million minutes	Verified for US East. Mumbai (ap-south-1) is not published on the main page and typically runs slightly higher — confirm in the AWS Pricing Calculator.
Hardware, television and advertising rates	Indian-market estimates

Sources

- MathWorks — ThingSpeak License Options: thingspeak.mathworks.com/prices
- MathWorks — ThingSpeak Standard and Academic licence pages
- Amazon Web Services — AWS IoT Core Pricing: aws.amazon.com/iot-core/pricing
- AWS IoT Core pricing documentation (developer guide)